

B.Tech. Degree V Semester Supplementary Examination
April 2019

ME 15-1502 MECHANICS OF MACHINERY
 (2015 Scheme)

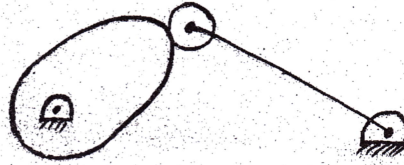
Time: 3 Hours

Maximum Marks: 60

PART A
 (Answer *ALL* questions)

(10 × 2 = 20)

- I. (a) Differentiate between lower and higher pairs.
 (b) Explain the mechanism of elliptical trammel and formulate the equation for ellipse.
 (c) Explain reverted gear train and its applications.
 (d) Differentiate between involute and cycloidal tooth profile.
 (e) Find the degree of freedom for the mechanism shown below.

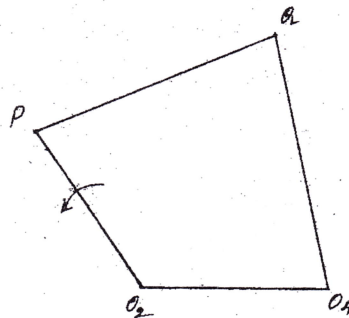


- (f) What are the different systems of gear teeth?
 (g) Differentiate between roller and mushroom followers.
 (h) Describe how a belt transmission dynamometer measures transmitted power.
 (i) Write the conditions for overhauling and self-locking of screws and explain it.
 (j) Differentiate between single plate clutch and multi-plate clutch.

PART B

(4 × 10 = 40)

- II. (a) Describe with suitable examples the inversions of a single slider crank mechanism. (4)
 (b) The input link O_2P of a four bar linkage is rotated at 2 rad/s in counter clockwise direction as shown below. Find the angular velocity of the coupler PQ in rad/s at an instant when angle $O_4O_2P = 120^\circ$. Take $PQ = O_4Q = \sqrt{2} a$, $O_2P = O_2O_4 = a$. Link O_2O_4 is fixed. (6)

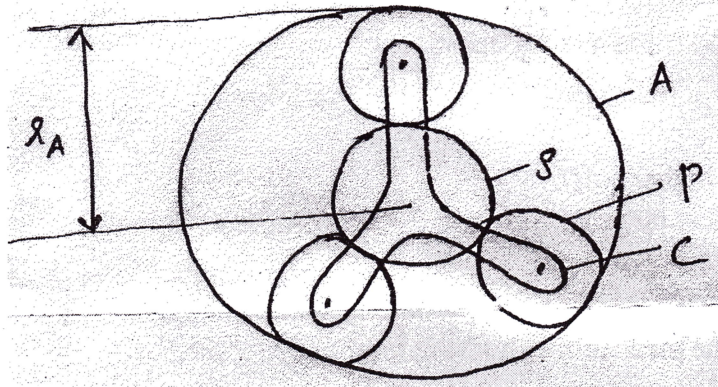


OR

- III. In a four bar chain ABCD, link AD is fixed and the crank AB rotates at 10 rad/s clockwise. Length of the links are $AB = 60$ mm, $BC = 70$ mm, $CD = 65$ mm and $AD = 120$ mm. When angle $DAB = 60^\circ$, both B and C lies on the same side of AD. Find (i) angular velocities of BC and CD (ii) angular accelerations of BC and CD. (10)

(P.T.O)

- IV. (a) Explain the different classification of gear drives. (4)
- (b) An epicyclic gear train of sun and planet type has a fixed outer annular wheel (A). The sun wheel (S) rotates at a speed of 600 rpm and the spider (arm C) carries three planet wheels (P). The number of teeth on the wheels S and P are $T_s = 18$ and $T_p = 27$. The diametrical pitch is same for all the mating gears. Find the number of teeth on wheel A and speed of wheel P. (6)



OR

- V. Two involute gears of 20° pressure angle are in mesh. The number of teeth on pinion is 20 and the gear ratio is 2. If the pitch expressed in module is 5 mm and the pitch line speed is 1.2 m/s, assuming the system is full depth involute, find (i) the angle turned through by pinion when one pair of teeth is in mesh, (ii) the maximum velocity of sliding. (10)

- VI. (a) Explain cam and follower mechanism and its classification. (4)
- (b) Design a cam profile having 100° out stroke, 60° dwell, 100° return stroke. The follower is radial type knife edge follower, stroke is 20 mm, basic circle radius is 25 mm, and the follower is in SHM. (6)

OR

- VII. The thrust of a propeller shaft in a marine engine is taken up by a number of collars integral with the shaft which is 300 mm in diameter. The thrust on the shaft is 200 kN and the speed is 75 rpm. Taking μ as constant and equal to 0.05 and assuming intensity of pressure as uniform and equal to 0.3 N/mm^2 , find the external diameter of the collars and the number of collars required, if the power lost in friction is not to exceed 16 kW. (10)

- VIII. A plate clutch has three discs on the driving shaft and two discs on the driven shaft, providing four pairs of contact surfaces. The outside diameter of the contact surfaces is 240 mm and inside diameter 120 mm. Assuming uniform pressure and $\mu = 0.3$; find the total spring load pressing the plates together to transmit 25 kW at 1575 rpm. If there are 6 springs each of stiffness 13 kN/m and each of the contact surfaces has worn away by 1.25 mm, find the maximum power that can be transmitted, assuming uniform wear. (10)

OR

- IX. (a) Determine the proportions of four bar mechanism, by using three precision points, to generate $y = x^{1.5}$, where x varies between 1 and 4. Assume $\theta_s = 30^\circ$; $\Delta\theta = 90^\circ$; $\Phi_s = 90^\circ$; and $\Delta\phi = 90^\circ$. Take length of the fixed link AD as 25 mm. (8)
- (b) What do you mean by synthesis of mechanism? Explain the various steps involved. (2)
